

Mapping Toronto's Crime Data – Tableau Story

“Mapping Toronto's Crime Data” is a Tableau Story Visualization Project I presented as my final project for Data Visualization course in the Big Data Analytics program at Georgian College. The aim of this project was to use Tableau for visual analysis, designing and storyboarding of readily available open data to convey insights in the form of a Data Story.



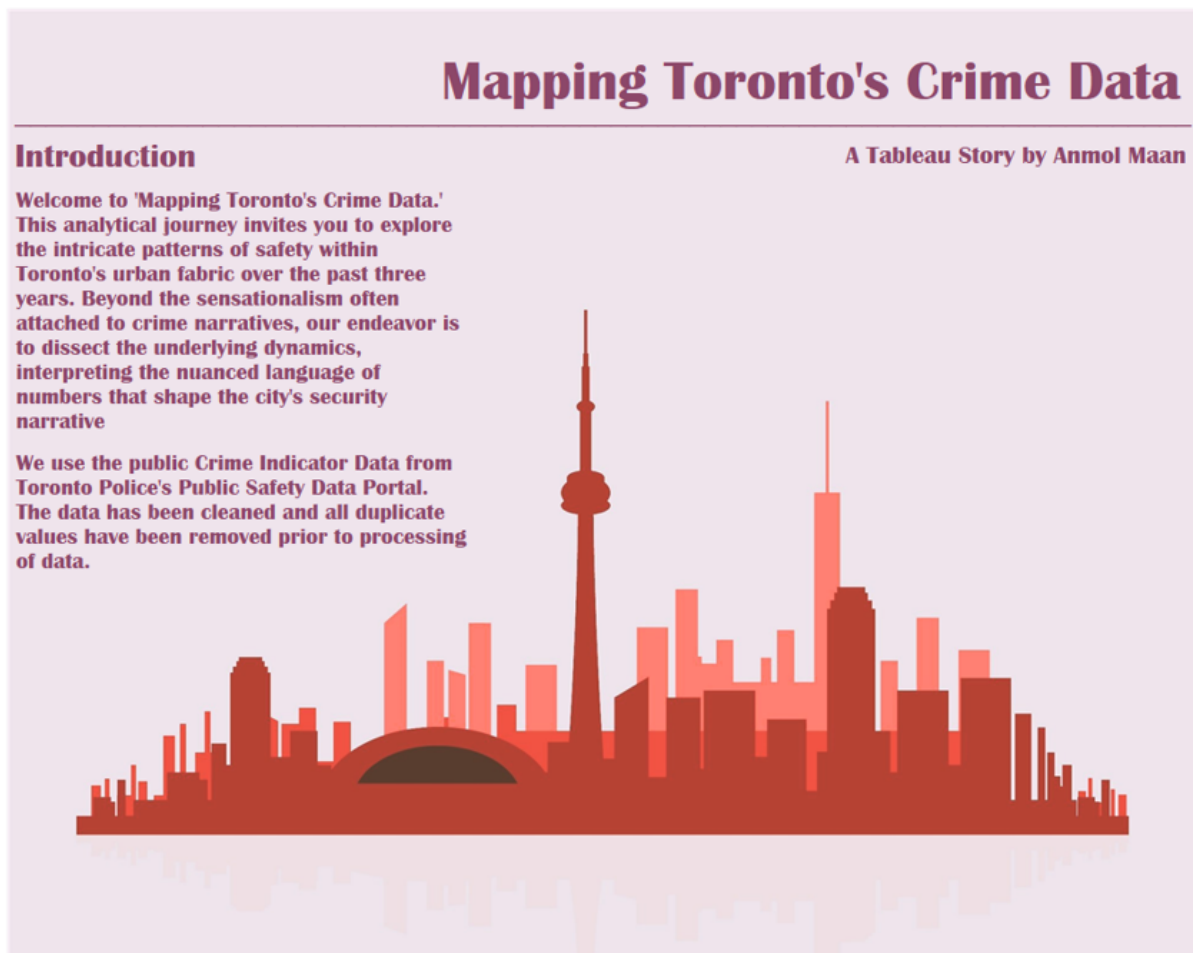
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Introduction

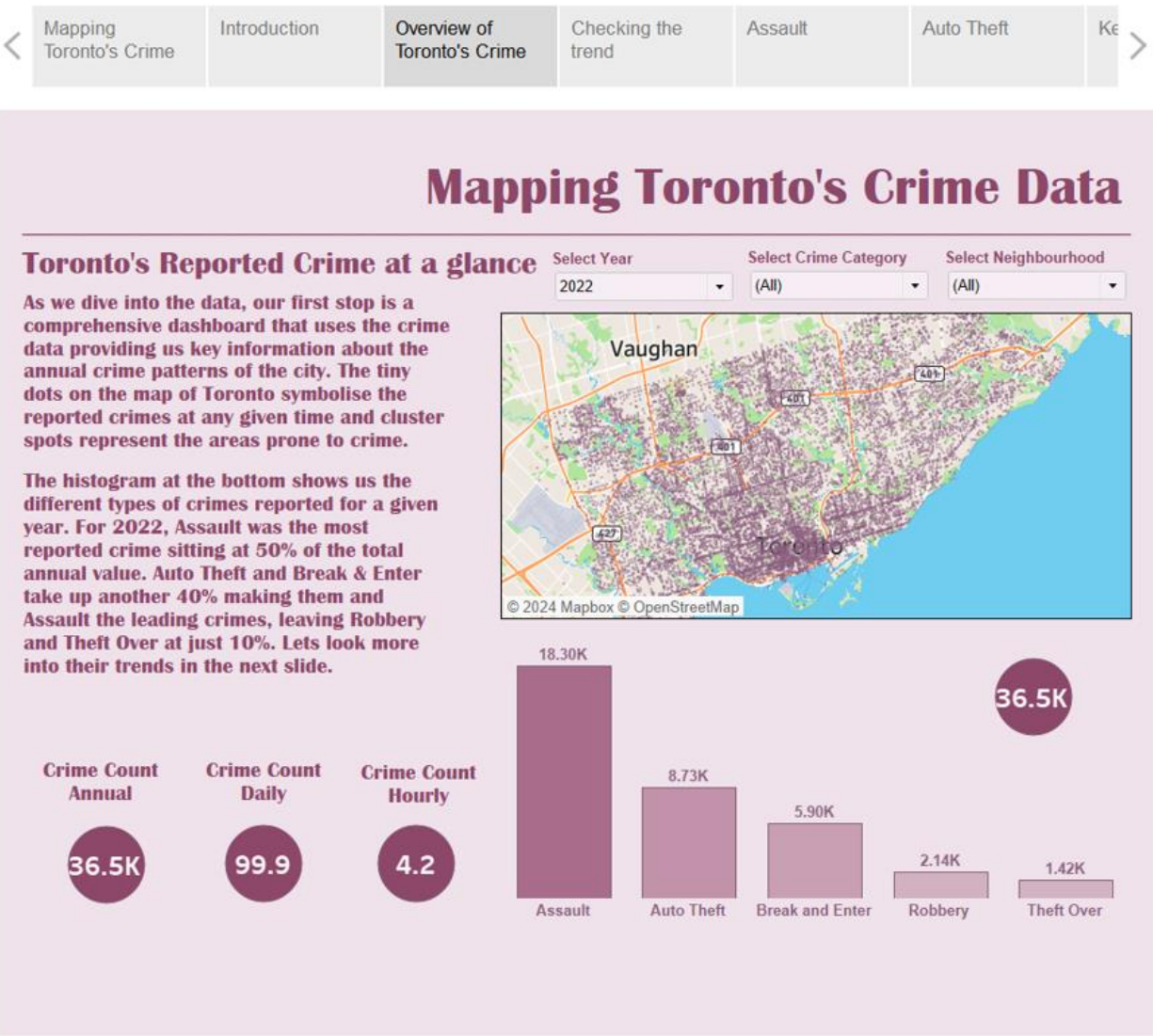
Finding something meaningful that would feed my curiosity was really tricky, but after lots of research, I selected Public Crime Indicators Data from Toronto Police's Public Safety Data Portal. The data ranged from 2014 to 2022 with over 300k records, which is reduced to the most recent three years for tracking trends and understanding how different neighbourhoods experienced different crimes. Excess data was removed to improve processing speeds after it is imported in MS Excel for initial exploration and cleaned of duplicates before visualizing in Tableau.

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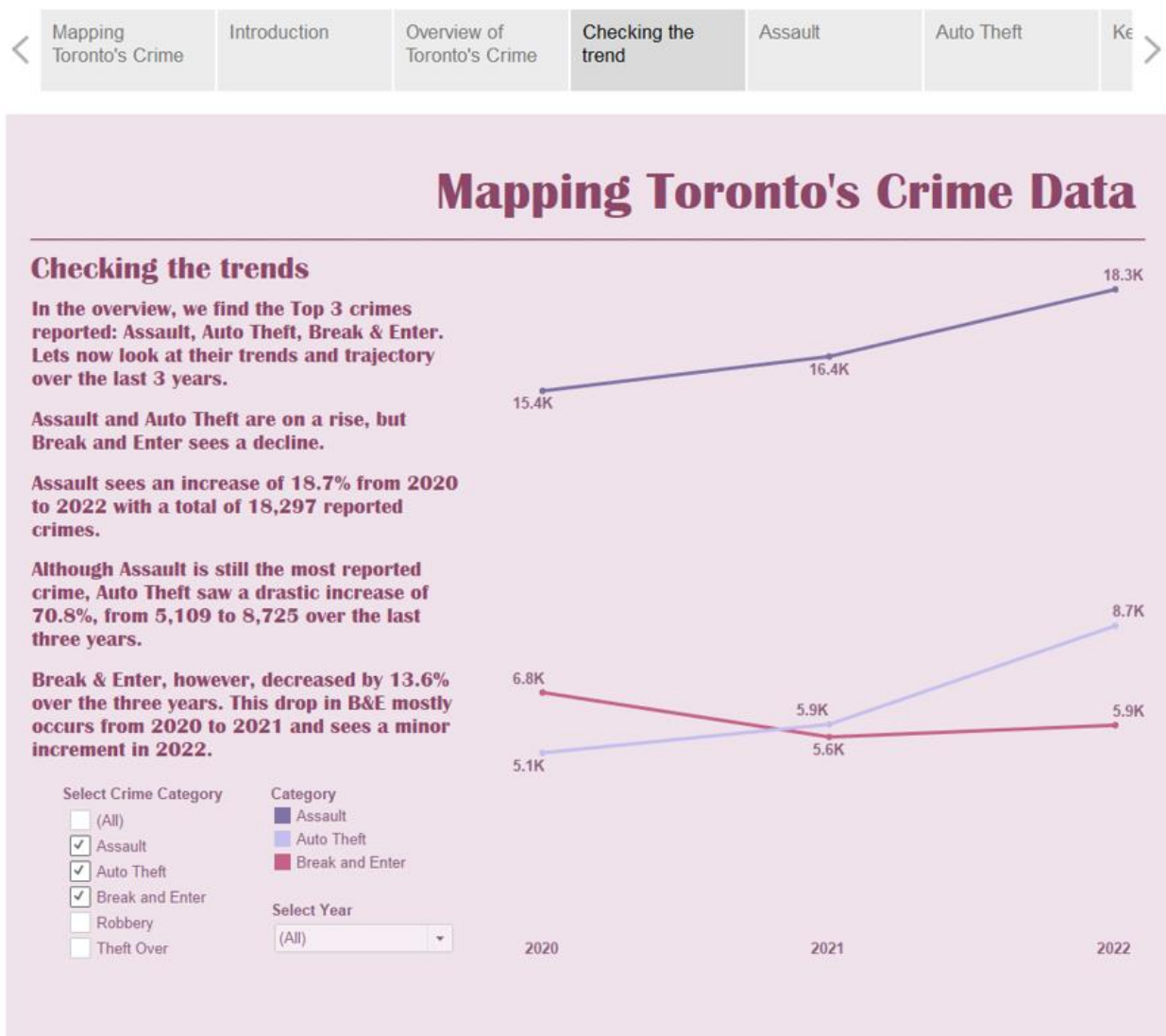
Overview of Toronto's Crime

In Tableau, I created a dashboard that provides a comprehensive overview of the annual crime patterns in the City of Toronto. It features a map representing reported crimes and cluster spots indicating high-crime areas. These can be further segregated to see the most affected neighbourhoods for each crime type using the dropdown filter or the histogram at the bottom. This histogram shows the distribution of different crime types for the selected year, highlighting that Assault was the most reported crime, followed by Auto Theft and Break & Enter in the most recent year, 2022. Accounting for around 90% of all reported crimes, these three crime types are the most prevalent in society. Let's dive deeper into the trends of these crimes and understand the trajectory of their distribution in recent years.



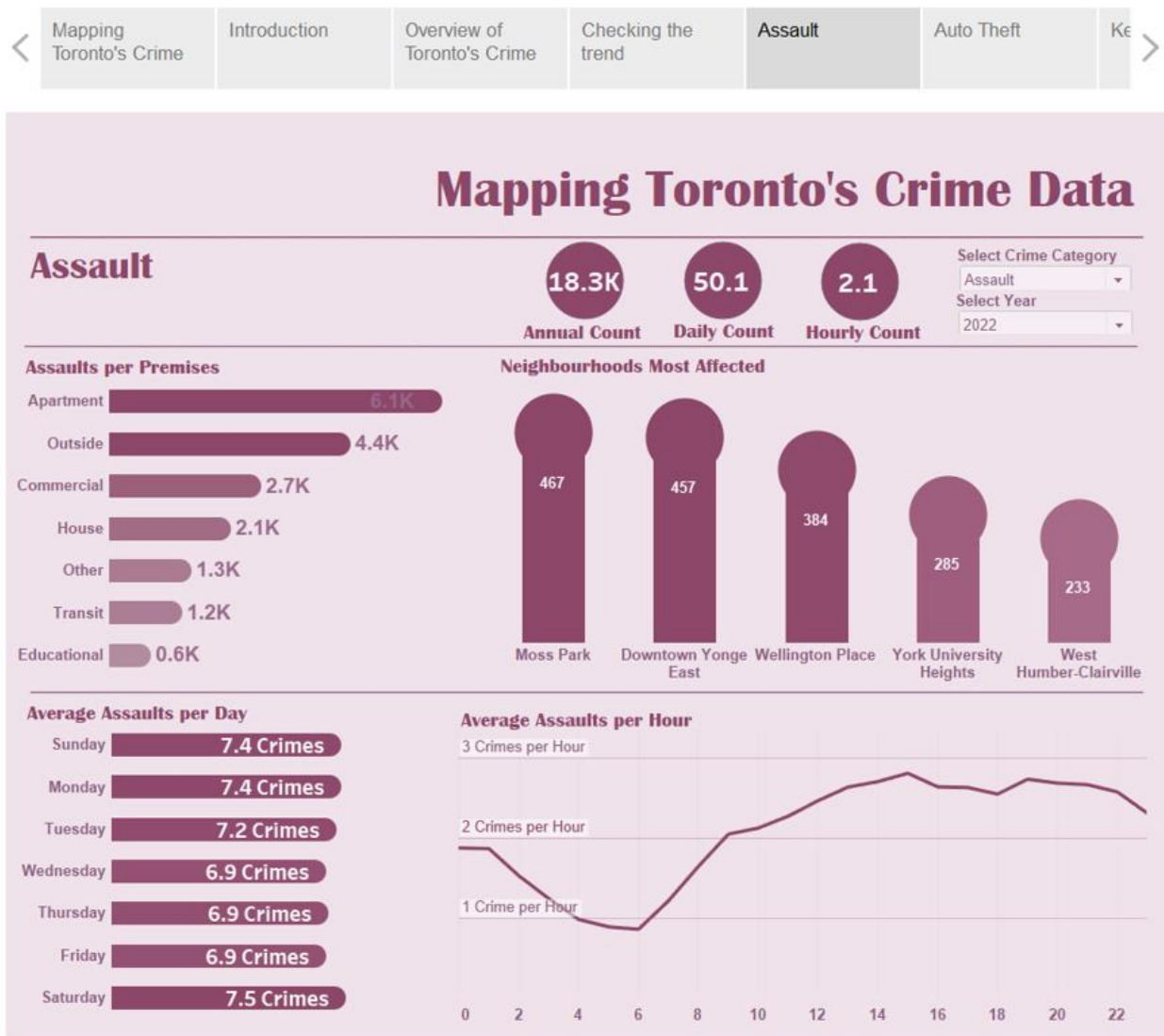
Checking the Trend

This section focuses on the trends and trajectories of the top three reported crimes: Assault, Auto Theft, and Break & Enter over the last three years. It notes that Assault and Auto Theft have been on the rise, with Auto Theft experiencing a significant increase of 70.8%. In contrast, Break & Enter has seen a decline of 13.6%, mostly from 2020 to 2021. This analysis helps in understanding the changing landscape of crime in Toronto and leads us to further examine the two crime types on the rise.



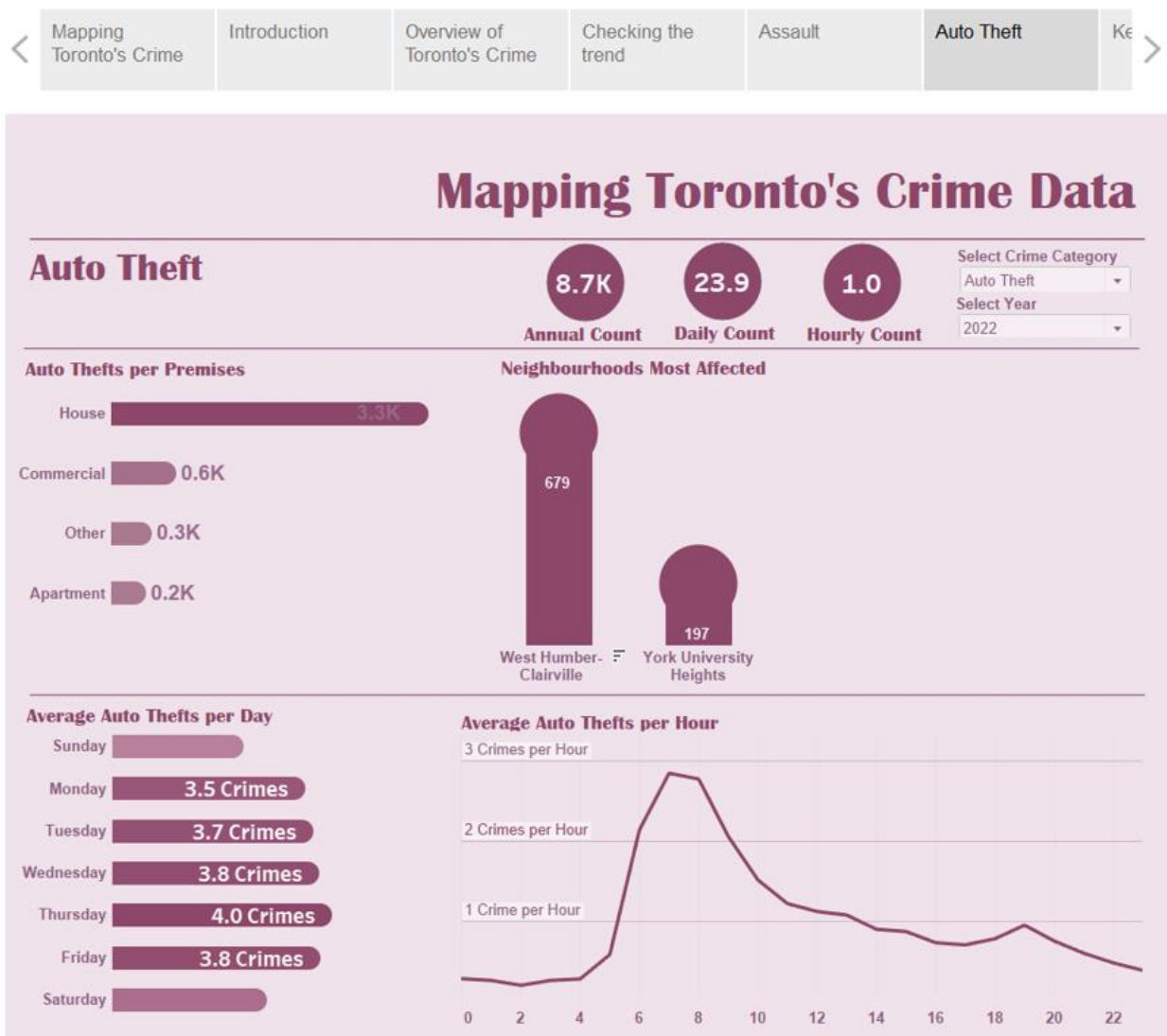
Assault

This story segment delves into the specifics of Assault crimes in Toronto. It provides detailed insights into the average number of assaults per day and per hour, with visualizations showing the frequency of assaults on different days of the week and in various types of premises. The neighborhoods most affected by Assault are also highlighted, offering a clear picture of where and when these crimes are most prevalent.



Auto Theft

This section provides an in-depth analysis of Auto Theft in Toronto. Similar to the Assault page, it includes visualizations of the average number of auto thefts per day and per hour, and identifies the most affected neighborhoods. It also breaks down the locations where auto thefts are most likely to occur, such as houses, commercial areas, and apartments, giving a comprehensive view of this crime category.



Key Takeaways – Is Toronto Safe?

The final section summarizes the key findings of the analysis. It notes that the annual crime count for 2022 was 36.5K, with Assault and Auto Theft being the top crime categories. It highlights the 70% increase in Auto Theft over the past three years and a decline in Break & Enter. The affected areas are identified, and the overall crime rate is contextualized with Toronto's population, suggesting that personal safety perceptions may vary depending on one's location and interactions.

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Mapping Toronto's Crime Data

Key Takeaways

Annual Crime Count for 2022 was 36.5k

Top Crime Categories were Assault and Auto Theft with B&E on decline and a 70% Increase over the three years in Auto Theft.

18.3k Annual Assault Reports and 8.7k Annual Auto Theft Reports.

Areas most affected with Assault and Auto Theft: Moss Park, Downtown Yonge East, Wellington Place, York Uni Heights, West Humber Clairville

Is Toronto Safe?

Is Toronto Safe?

Population of Toronto in 2022 was 3.025M

Total Annual crime reports/Population comes to about 1.2%

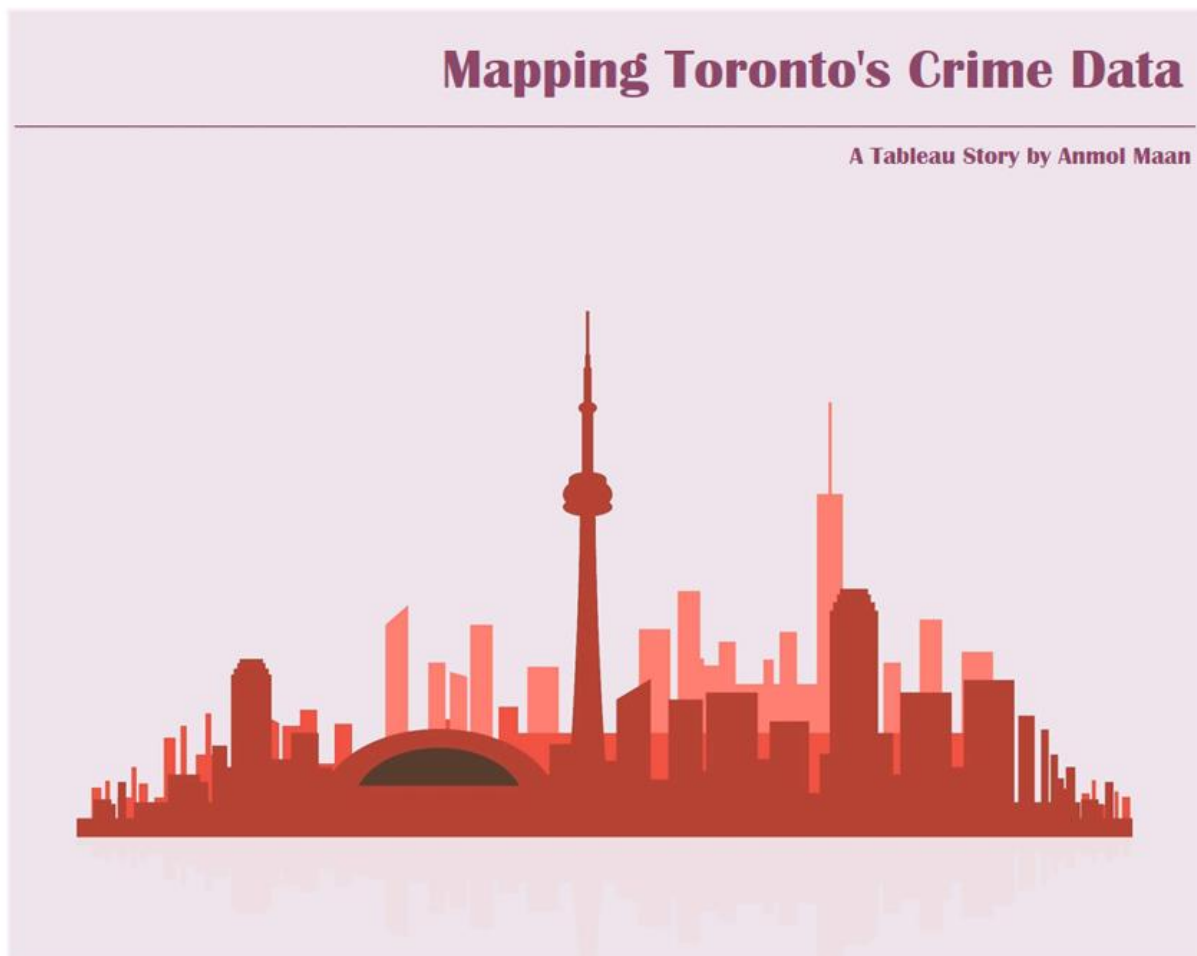
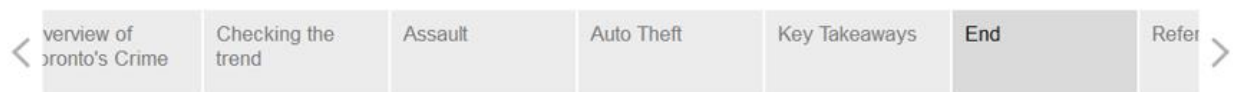
Whether you find it safe or not depends on where you live and your interactions with others.

Always be aware of your surroundings.

Stay Safe!
Anmol Maan

End

In conclusion, by focusing on key crime types such as Assault and Auto Theft, this project highlights the changing trends and the most affected neighborhoods over the last three years. The visualizations created, including maps, histograms, and detailed dashboards, provide a comprehensive understanding of the crime landscape in Toronto. This analysis not only aids in recognizing patterns but also serves as a valuable tool for public awareness and safety planning. Through this project, I have demonstrated the power of data visualization in uncovering important societal trends and contributing to informed decision-making.



References

This page lists the sources of data and illustrations used in the Tableau story. It provides the URLs for the Toronto Police's Public Safety Data Portal and the illustration of Toronto's skyline, ensuring transparency and credibility of the information presented in the analysis.

- Toronto Police Service. (n.d.). Major Crime Indicators Open Data. Retrieved from <https://data.torontopolice.on.ca/datasets/TorontoPS::major-crime-indicators-open-data/about>
- Freepik. (n.d.). Red Toronto Skyline Illustration. Retrieved from https://www.freepik.com/free-vector/red-toronto-skyline_1797690.htm#query=toronto&position=0&from_view=keyword&track=sph&uuid=6fbaa006-2c9d-4298-9f44-487d3c70ef15



Mapping Toronto's Crime Data

References

A Tableau Story by Anmol Maan

Data retrieved from: <https://data.torontopolice.on.ca/datasets/TorontoPS::major-crime-indicators-open-data/about>

Toronto Illustration: https://www.freepik.com/free-vector/red-toronto-skyline_1797690.htm#query=toronto&position=0&from_view=keyword&track=sph&uuid=6fbaa006-2c9d-4298-9f44-487d3c70ef15

